

Shay Tanne

ML Engineer

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- ML Engineer with **4 years of experience in autonomous driving, ML systems, and performance analysis.**
- Building **ML evaluation, analysis, and monitoring systems** for production perception systems.
- Pairing solid **algorithmic foundations** with strong **software engineering know-how** to develop production-grade analytical workflows.
- **MSc in Computer Science** (ongoing) focusing on NLP & RL with emphasis on real-world applications.
- **Curious and proactive team player**, someone who goes the extra mile to **make things work**.

Skills & Proficiencies

ML & Algorithms: Reinforcement Learning • NLP • Deep Learning • Algorithm Development • Classical ML • Representation Learning & Mechanistic Interpretability • Computer Vision • Performance Analysis • Experiment Design • Statistical Analysis • Feature Engineering • Model Evaluation

SW System Engineering: Large-Scale Evaluation Pipelines • Performance & Regression Monitoring • API Design • Databases • Full-Stack Development • Microservices Architecture • Caching & Optimization • Cloud-Native Applications

Tools & Frameworks: PyTorch • TensorFlow • Hugging Face • NumPy • Pandas • Scikit-learn • SciPy • OpenCV • Weights & Biases • Apache Iceberg • Azure Data Explorer • Flask • FastAPI • Dash • Plotly • Kubernetes • Docker • Jenkins • Git • Linux

Soft Skills: Scrum Methodology • Technical Documentation & Communication • Data Visualization

Programming Languages: Python • JavaScript • HTML • SQL/KQL • C/C++ • Groovy

Experience

2022-2025 Software/Algorithm Engineer (Ultra-Cruise AV), General Motors

- Building **algorithmic analysis systems** for **autonomous vehicle perception models**, enabling deep investigation of their behavior and failure modes.
- Designing **interactive and automated analysis workflows** to increase observability & validation coverage of system functions and improve performance measurability.
- Developing an algorithmic engine for **scalable analysis** of large simulation/recording datasets.
- Implementing **performance monitoring pipelines** to detect regression **before** deployment.
- Contributing tools that led to **~2x rise in discovered** perception bugs, improving delivery quality.
- **Optimizing** computational workflows with vectorization, caching, and predictive analysis.
- Driving **full development cycle** of new features and workflows – requirement gathering & research, design, implementation, testing, deployment, and integration.

2020-2021 Software Developer, UBC Program in Cryo-EM

- **Building production engineering backbone** for pharmaceutical research (**published in *Science and Nature Communications***) – accelerated data collection bottleneck processes by 4–10x.
- Developing and deploying **CNN-based models** for molecular image classification and ranking.
- Designing **data processing, training, and evaluation workflows** for large-scale datasets.

2019 Software Engineering Intern, Microsoft Innovation Lab

- Collaborating with product teams and external partners across healthcare, aviation, and environmental domains on highly innovative, small-scale experimental projects.
- Concurrent work on several projects utilizing varying core technologies (ML, CV, 3D modeling) to develop novel solutions to customer pain-points.
- Management and hands-on development of new product prototypes, from ideation to PoC.

Personal/Research Projects

2026 (MSc) NLP Research - Geometry of Factuality in LLM Internals

- Leading an **end-to-end LLM interpretability research** project analyzing latent representation geometry to detect falsehoods at inference-time using various trained LLM probes.
- **Engineering versatile experimentation infrastructure with tools and pipelines** for running experiment sweeps, analyzing results, and generating & managing artifacts.
- **Defining research questions & hypotheses**, constructing methodologies, **designing & implementing experiments**, analyzing results, and **drawing scientific conclusions**.
- Utilizing techniques from linear algebra, statistics, and classical ML to ensure scientific rigor, control confounding features, enforce data hygiene, and cross-validate results.

2025 (MSc) RL Research Project - Multi-Task Navigation Environments

- **Designing and training RL agents** to solve multi-task MiniGrid environments with sparse rewards.
- **Implementing** and methodically **analyzing several algorithmic approaches**: DDQN+PER vs A2C vs PPO+GAE, reward shaping, and hyperparameter tuning.
- **Developing experiment pipelines** and results **processing tools** using Weights & Biases.
- Running controlled experiments to analyze **sample efficiency, convergence, and policy stability**.

Education

2025-present MSc Computer Science, Reichman University

Thesis-based degree, focusing on ML with emphasis on modern NLP and RL.

2017-2021 BSc Electrical Engineering, University of British Columbia

Areas of focus: ML and data science, signal processing, complex SW systems.

Military Service & Volunteering

2011-2015 Infantry Officer

- Deputy Captain – 2nd in command of an infantry company (120 personnel).
- Infantry Team Head – command in training and deployment (30 personnel, Operation Tzuk Eitan).

2009-2016 Youth Movement Counselor, Manager, Journal Head Editor (including Shnat-Sherut)

Extras

Languages: English (native), Hebrew (native).

Extracurriculars: Long distance running (ultramarathon), mountaineering, ski touring, playing music.